

Econometrics I

Lecture 5b: Causal Diagrams (DAGs)

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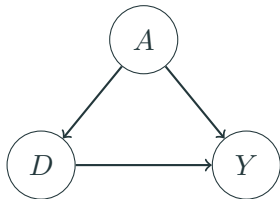
NYU Stern

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Two languages for the same problem

- ▶ **Potential outcomes** (last hour) tell us *what we want*: $\mathbb{E}[Y_i(1) - Y_i(0)]$, and what randomization buys us.
- ▶ **Causal diagrams** (DAGs) encode *what we believe*: which variables cause which, drawn as arrows. The assumptions are the *missing* arrows.
- ▶ Why learn both?
 - DAGs make “what should I control for?” a mechanical question with a right answer — and expose when a control makes things **worse**.
 - Potential outcomes handle heterogeneity and identification-by-design more gracefully. Economists mostly publish in this language.
- ▶ Readings: Mixtape Ch. 3; Hünermund & Bareinboim (2021); Cinelli, Forney & Pearl (2022), “A Crash Course in Good and Bad Controls.”

What is a DAG?



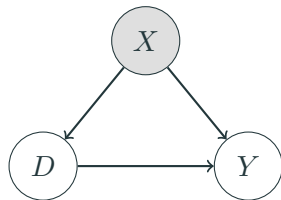
D = education, Y = wages, A = ability

- ▶ Nodes are variables; arrows are direct causal effects; no cycles allowed (“acyclic”).
- ▶ Every **missing arrow is an assumption**: this graph asserts nothing else causes both D and Y .
- ▶ Unobserved variables (like ability) still belong in the graph — being unobservable doesn’t stop something from causing things.

In the graph on the last slide there are two paths from D to Y :

1. $D \rightarrow Y$: the **causal path**. This is the thing we want to measure.
 2. $D \leftarrow A \rightarrow Y$: a **backdoor path**. Association flows along it even though D causes none of it.
- ▶ The regression of Y on D alone picks up *both* paths — this is exactly the omitted variables bias formula from Lecture 2, drawn as a picture.
 - ▶ **Identification strategy** = a plan for shutting every backdoor path while leaving the causal path alone.

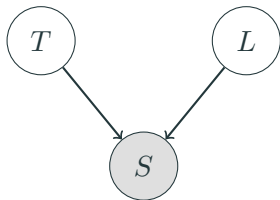
Good controls: closing a backdoor



Shading = conditioned on.

- Conditioning on a **confounder** X (regression control, matching, stratification) **blocks** the backdoor path $D \leftarrow X \rightarrow Y$.
- This is the **backdoor criterion** (informally): find a set of observed variables that blocks every backdoor path, condition on them, and the remaining association is causal.
- “Selection on observables” (matching, next month) is exactly the claim that such a set exists *and you have it*.

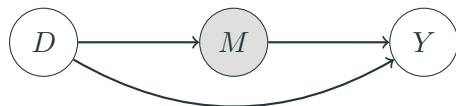
Colliders: conditioning can open a path



T = talent, L = luck, S = startup gets funded.

- ▶ S is a **collider** on the path $T \rightarrow S \leftarrow L$: two arrows collide into it. A collider **blocks** the path by default.
- ▶ But **conditioning on a collider opens the path**: among *funded* startups, talent and luck are negatively correlated — if a funded founder wasn't talented, they must have been lucky.
- ▶ Talent and luck are independent in the population. The correlation is manufactured by the conditioning. This is **selection bias, drawn as a picture**.

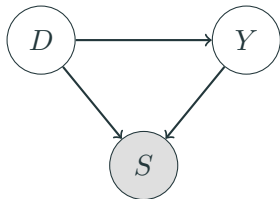
Bad control #1: the post-treatment variable



D = education, M = occupation, Y = wages

- ▶ M is a **mediator**: part of *how* education raises wages is by changing occupations.
- ▶ Controlling for occupation **blocks part of the effect you are trying to measure**. The coefficient on D is no longer the total effect.
- ▶ Rule of thumb with real teeth: **do not control for variables that are themselves outcomes of the treatment**. When in doubt, ask: was this variable determined before or after D ?

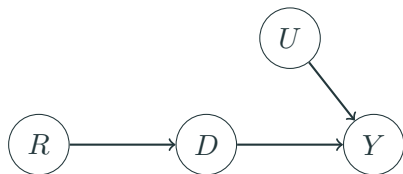
Bad control #2: the collider / sample-selection control



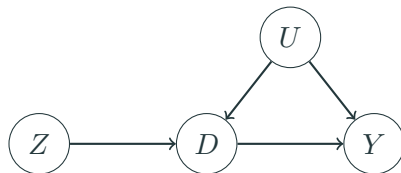
D = ad exposure, Y = purchase intent, S = visits
the site

- ▶ Estimating the ad effect **only among site visitors** conditions on a collider: both seeing the ad and wanting the product cause visits.
- ▶ Within visitors, exposure and intent are spuriously related even if the ad does nothing.
- ▶ This is the same disease as Heckman's sample selection (Lecture 11) and the "conditioning on graduation" problem in education studies. **Who is in your sample, and did the treatment help put them there?**

The research designs of this course, as DAGs



Randomization: coin flip R causes D ; no arrow into D from U .



Instrumental variables: the missing arrows $Z \rightarrow Y$ (exclusion) and $U \rightarrow Z$ (exogeneity) ARE the IV assumptions.

Every design we meet from here on is a claim about missing arrows. When you referee a paper, **draw its DAG** — the contestable assumption is usually an arrow the authors need to be absent.

Where DAGs strain (and economists push back)

- ▶ **Simultaneity**: supply and demand determine price and quantity *jointly* — a cycle, which a DAG cannot draw. Our oldest identification problem (Working 1927, Lecture 6) sits awkwardly in this language.
- ▶ **Heterogeneous effects**: LATE, compliers, and monotonicity (Lecture 7) are natural in potential outcomes and clumsy in DAGs.
- ▶ **Where DAGs shine**: bad controls, sample selection, and transparent communication of assumptions — especially in fields (marketing, IS) where “we controlled for everything” is still treated as a virtue.
- ▶ **Verdict**: use the DAG to decide *what to control for*; use potential outcomes to decide *what your estimate means*.

Coming attraction: when units interfere

- ▶ Everything today assumed my treatment doesn't affect *your* outcome (part of **SUTVA**). Drawn as a DAG: no arrows *between units*.
- ▶ In marketplaces and platforms this fails constantly:
 - Discounting rides for treated riders lengthens wait times for control riders.
 - Promoting treated sellers steals demand from control sellers $\Rightarrow$ the A/B test **overstates** the effect of the promotion.
- ▶ Fixes people use: cluster-randomize markets, switchback designs, exposure modeling. See Wager Chs. 11–12 — and Econometrics II if you want the full treatment.
- ▶ For now: when someone shows you an experiment on a **connected** system, ask what the control group's counterfactual really was.

Recap

- ▶ A DAG is a set of causal assumptions; the missing arrows do the work.
- ▶ Backdoor paths are confounding; blocking them is what “controlling for” means.
- ▶ Colliders reverse the logic: conditioning *creates* association. Selection into the sample is a collider.
- ▶ Bad controls: post-treatment variables and colliders. Ask *when* the variable was determined and *whether treatment helped put units in the sample*.
- ▶ Every design in this course is a claim about missing arrows — draw the picture before you believe the number.

Let's look at the workbook now.

- ▶ Open `inclass_session5.Rmd` (in the L7 - Program Evaluation and Selection folder of the course repo).
- ▶ Reproduce the power calculations and the funded-startup collider.
- ▶ Then work the two problems with a neighbor:
 1. The bad control, live
 2. Design your own experiment

Thanks!
